

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

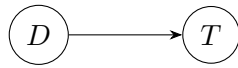
1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

Interpretation. $D = 1$: the patient has the disease. $T = 1$: the test result is positive.

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

Idea. The disease causes the test result. The causal graph has one arrow:



Write the goal as a ratio, and find the numerator and the denominator from the joint distribution:

$$\begin{aligned} p(D \mid T) &= \frac{p(D, T)}{p(T)}, \\ p(D, T) &= p(D) p(T \mid D) && \text{(chain rule; both factors are given),} \\ p(T) &= \sum_d p(D = d, T) && \text{(sum over the parent } D). \end{aligned}$$

Solution.

Step 1. Write the joint distribution.

$$p(D, T) = p(D) p(T \mid D) \quad \text{(chain rule; parent of } T: D).$$

Step 2. Write the goal as a ratio.

$$p(D = 1 \mid T = 1) = \frac{p(D = 1, T = 1)}{p(T = 1)}.$$

Step 3. Find the numerator. All variables are fixed, so there is no sum:

$$p(D = 1, T = 1) = p(D = 1) p(T = 1 \mid D = 1) = 0.0001 \cdot 0.99 = 0.000099.$$

Step 4. Find the denominator. Sum the joint distribution over D :

$$\begin{aligned} p(T = 1) &= \sum_d p(D = d) p(T = 1 \mid D = d) \\ &= p(D = 1) p(T = 1 \mid D = 1) + p(D = 0) p(T = 1 \mid D = 0) \quad (d \in \{0, 1\}) \\ &= 0.000099 + (1 - p(D = 1)) (1 - p(T = 0 \mid D = 0)) \quad \text{(Step 3; complements)} \\ &= 0.000099 + (1 - 0.0001) \cdot (1 - 0.99) \\ &= 0.000099 + 0.9999 \cdot 0.01 \\ &= 0.000099 + 0.009999 = 0.010098. \end{aligned}$$

Step 5. Divide.

$$\begin{aligned}
 p(D = 1 \mid T = 1) &= \frac{0.000099}{0.010098} && \text{(Steps 3, 4)} \\
 &= \frac{99}{10098} && \text{(multiply by } 10^6) \\
 &= \frac{99}{99 \cdot 102} && (10098 = 99 \cdot 102) \\
 &= \frac{1}{102} \approx 0.0098.
 \end{aligned}$$

Conclusion. The probability increases from 0.01% to only 1%. In 10 000 people, the test finds $10\,000 \cdot 0.000099 \approx 1$ sick person and $10\,000 \cdot 0.009999 \approx 100$ healthy people. Thus most positive results of a mass test for a rare disease are false. A second positive independent test uses $p(D = 1 \mid T_1 = 1) = 1/102$ as the prior:

$$\begin{aligned}
 p(D = 1 \mid T_1 = 1, T_2 = 1) &= \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + (1 - \frac{1}{102}) \cdot 0.01} && \text{(Bayes' theorem)} \\
 &= \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \frac{101}{102} \cdot 0.01} && (1 - \frac{1}{102} = \frac{101}{102}) \\
 &= \frac{0.99}{0.99 + 101 \cdot 0.01} && \text{(multiply by 102)} \\
 &= \frac{0.99}{0.99 + 1.01} = \frac{0.99}{2} = 0.495.
 \end{aligned}$$

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D)$	P	D	$p(H = 1 \mid P)$	P	$p(T = 1 \mid A)$	A
0.999	1	1	0.9	1	0.95	1
0.9	1	0	0.2	0	0.5	0
0.9	0	1				
0.01	0	0				

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

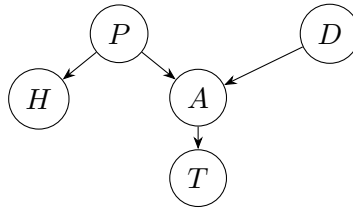


Figure 1: Causal relations for Problem 2

Idea. Problem 1 is a special case of this method.

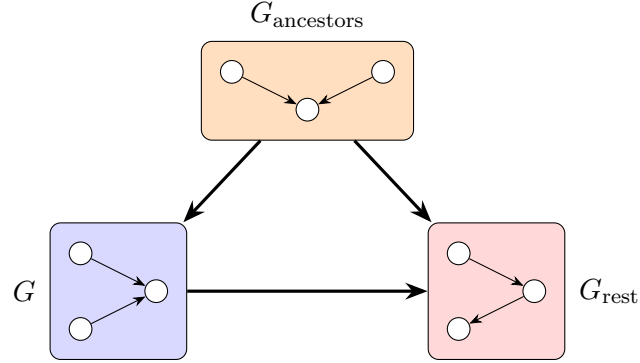
Joint distribution. The problem gives the distribution of each source node and the distribution of each other node given its parents. The chain rule, from the children to the parents, gives the joint distribution as the product of these factors. Example for the graph $X \rightarrow Z \leftarrow Y$:

$$\begin{aligned}
p(X, Y, Z) &= p(X, Y) p(Z \mid X, Y) && \text{(chain rule)} \\
&= p(X) p(Y \mid X) p(Z \mid X, Y) && \text{(chain rule)} \\
&= p(X) p(Y) p(Z \mid X, Y) && (Y \text{ has no parents}).
\end{aligned}$$

Algorithm for $p(G)$, where G is a group of variables with fixed values:

1. $G_{\text{ancestors}}$: the ancestors of G that are not in G . G_{rest} : all other variables.
2. $p(G) = \sum_{G_{\text{ancestors}}} p(G, G_{\text{ancestors}})$: the sum over G_{rest} is 1.
3. G and $G_{\text{ancestors}}$ together contain the parents of all their variables. Thus the chain rule gives $p(G, G_{\text{ancestors}})$ as a product of the given tables.
4. Move the factors that do not depend on a summation variable out of its sum. Then calculate.

Proof of Steps 2 and 3. No arrow goes from G_{rest} to G or $G_{\text{ancestors}}$: otherwise, its start is an ancestor of G . Thus the parents of each variable in $G \cup G_{\text{ancestors}}$ are in $G \cup G_{\text{ancestors}}$:



Write the joint distribution as the product of the factors $p(v \mid \text{parents of } v)$, and split it into two products:

$$\begin{aligned}
p(G) &= \sum_{G_{\text{ancestors}}} \sum_{G_{\text{rest}}} \prod_{v \in G \cup G_{\text{ancestors}}} p(v \mid \text{parents of } v) \prod_{r \in G_{\text{rest}}} p(r \mid \text{parents of } r) \\
&= \sum_{G_{\text{ancestors}}} \prod_{v \in G \cup G_{\text{ancestors}}} p(v \mid \text{parents of } v) \sum_{G_{\text{rest}}} \prod_{r \in G_{\text{rest}}} p(r \mid \text{parents of } r) \quad (\text{constants out}) \\
&= \sum_{G_{\text{ancestors}}} \prod_{v \in G \cup G_{\text{ancestors}}} p(v \mid \text{parents of } v) \quad (\text{sum is 1}).
\end{aligned}$$

The first product does not contain G_{rest} , because the parents of $G \cup G_{\text{ancestors}}$ are in $G \cup G_{\text{ancestors}}$. The last sum is 1: sum first over a variable of G_{rest} without children in G_{rest} . Its factor sums to 1, and no other factor contains it. Then repeat.

For a query Q and an observation E , find $p(Q, E)$ and $p(E)$ with the algorithm, then divide:

$$p(Q \mid E) = \frac{p(Q, E)}{p(E)}.$$

Solution. The tables give $p(X = 1 \mid \text{parents})$ for each node X , and $p(X = 0 \mid \text{parents}) = 1 - p(X = 1 \mid \text{parents})$.

Joint distribution. Apply the chain rule one variable at a time, from the children to the parents. A missing arrow means no direct influence: given its parents, a node does not depend on the

other nodes in the condition.

$$\begin{aligned}
p(P, D, A, H, T) &= p(P, D, A, H) p(T \mid P, D, A, H) && \text{(chain rule)} \\
&= p(P, D, A, H) p(T \mid A) && \text{(parent of } T: A) \\
p(P, D, A, H) &= p(P, D, A) p(H \mid P, D, A) && \text{(chain rule)} \\
&= p(P, D, A) p(H \mid P) && \text{(parent of } H: P) \\
p(P, D, A) &= p(P, D) p(A \mid P, D) && \text{(chain rule; parents of } A: P, D) \\
p(P, D) &= p(P) p(D \mid P) && \text{(chain rule)} \\
&= p(P) p(D) && (D \text{ has no parents})
\end{aligned}$$

Thus

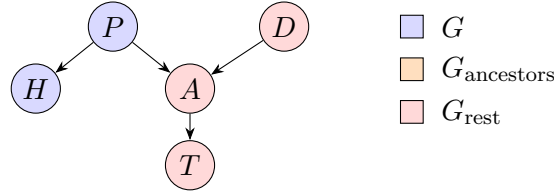
$$p(P, D, A, H, T) = p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A).$$

(a) Find $p(P = 1 \mid H = 1)$.

Step 1. Write the goal as a ratio.

$$p(P = 1 \mid H = 1) = \frac{p(P = 1, H = 1)}{p(H = 1)}.$$

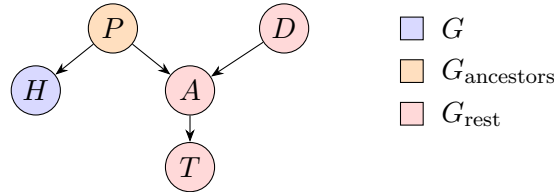
Step 2. Find the numerator: $G = \{P, H\}$, $G_{\text{ancestors}} = \emptyset$.



Joint distribution:

$$\begin{aligned}
p(\overline{P}, \overline{D}, \overline{A}, \overline{H}, \overline{T}) &= p(\overline{P}) p(\overline{D}) p(\overline{A} \mid \overline{P}, \overline{D}) p(\overline{H} \mid \overline{P}) p(\overline{T} \mid \overline{A}). \\
p(P = 1, H = 1) &= p(P = 1) p(H = 1 \mid P = 1) = 0.2 \cdot 0.9 = 0.18.
\end{aligned}$$

Step 3. Find the denominator: $G = \{H\}$, $G_{\text{ancestors}} = \{P\}$.



Joint distribution:

$$\begin{aligned}
p(\overline{P}, \overline{D}, \overline{A}, \overline{H}, \overline{T}) &= p(\overline{P}) p(\overline{D}) p(\overline{A} \mid \overline{P}, \overline{D}) p(\overline{H} \mid \overline{P}) p(\overline{T} \mid \overline{A}). \\
p(H = 1) &= \sum_p p(P = p) p(H = 1 \mid P = p) \\
&= 0.2 \cdot 0.9 + (1 - 0.2) \cdot 0.2 = 0.18 + 0.16 = 0.34.
\end{aligned}$$

Step 4. Divide.

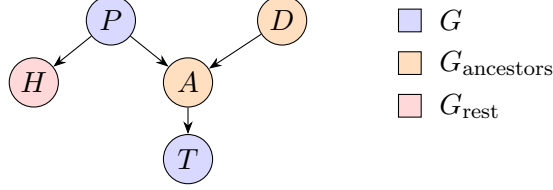
$$p(P = 1 \mid H = 1) = \frac{0.18}{0.34} = \frac{9}{17} \approx 0.529.$$

(b) **Find** $p(P = 1 \mid T = 1)$.

Step 1. Write the goal as a ratio.

$$p(P = 1 \mid T = 1) = \frac{p(P = 1, T = 1)}{p(T = 1)}.$$

Step 2. Find the numerator: $G = \{P, T\}$, $G_{\text{ancestors}} = \{A, D\}$.

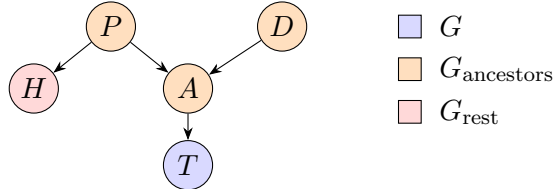


Joint distribution:

$$p(\textcolor{blue}{P}, \textcolor{orange}{D}, \textcolor{orange}{A}, \textcolor{pink}{H}, \textcolor{blue}{T}) = p(\textcolor{blue}{P})p(\textcolor{orange}{D})p(\textcolor{orange}{A} \mid \textcolor{blue}{P}, \textcolor{orange}{D})p(\textcolor{pink}{H} \mid \textcolor{blue}{P})p(\textcolor{blue}{T} \mid \textcolor{orange}{A}).$$

$$\begin{aligned} & p(P = 1, T = 1) \\ &= \sum_a \sum_d p(P = 1) p(D = d) p(A = a \mid P = 1, D = d) p(T = 1 \mid A = a) \quad (\text{algorithm}) \\ &= p(P = 1) \sum_a p(T = 1 \mid A = a) \sum_d p(A = a \mid P = 1, D = d) p(D = d) \quad (\text{constants out}) \\ &= 0.2 [0.95 (0.999 \cdot 0.4 + 0.9 \cdot (1 - 0.4)) \\ &\quad + 0.5 ((1 - 0.999) \cdot 0.4 + (1 - 0.9) \cdot (1 - 0.4))] \quad (a = 1, a = 0) \\ &= 0.2 [0.95 \cdot 0.9396 + 0.5 \cdot 0.0604] \\ &= 0.2 [0.89262 + 0.0302] = 0.2 \cdot 0.92282 = 0.184564. \end{aligned}$$

Step 3. Find the denominator: $G = \{T\}$, $G_{\text{ancestors}} = \{P, A, D\}$.



Joint distribution:

$$p(\textcolor{orange}{P}, \textcolor{orange}{D}, \textcolor{orange}{A}, \textcolor{pink}{H}, \textcolor{blue}{T}) = p(\textcolor{orange}{P})p(\textcolor{orange}{D})p(\textcolor{orange}{A} \mid \textcolor{orange}{P}, \textcolor{orange}{D})p(\textcolor{pink}{H} \mid \textcolor{orange}{P})p(\textcolor{blue}{T} \mid \textcolor{orange}{A}).$$

$$\begin{aligned} & p(T = 1) \\ &= \sum_p \sum_a \sum_d p(P = p) p(D = d) p(A = a \mid P = p, D = d) p(T = 1 \mid A = a) \quad (\text{algorithm}) \\ &= \sum_p p(P = p) \sum_a p(T = 1 \mid A = a) \sum_d p(A = a \mid P = p, D = d) p(D = d) \quad (\text{constants out}) \\ &= 0.184564 + (1 - 0.2) [0.95 (0.9 \cdot 0.4 + 0.01 \cdot 0.6) \\ &\quad + 0.5 ((1 - 0.9) \cdot 0.4 + (1 - 0.01) \cdot 0.6)] \quad (p = 1: \text{Step 2}) \\ &= 0.184564 + 0.8 [0.95 \cdot 0.366 + 0.5 \cdot 0.634] \\ &= 0.184564 + 0.8 [0.3477 + 0.317] = 0.184564 + 0.8 \cdot 0.6647 \\ &= 0.184564 + 0.53176 = 0.716324. \end{aligned}$$

Step 4. Divide.

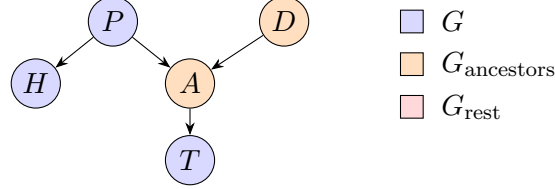
$$p(P = 1 \mid T = 1) = \frac{0.184564}{0.716324} \approx 0.258.$$

(c) Find $p(P = 1 \mid T = 1, H = 1)$.

Step 1. Write the goal as a ratio.

$$p(P = 1 \mid T = 1, H = 1) = \frac{p(P = 1, H = 1, T = 1)}{p(H = 1, T = 1)}.$$

Step 2. Find the numerator: $G = \{P, H, T\}$, $G_{\text{ancestors}} = \{A, D\}$.

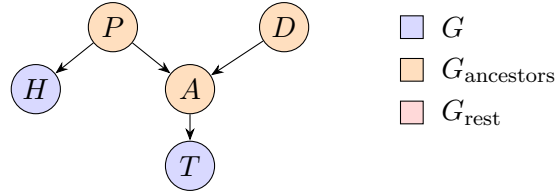


Joint distribution:

$$p(\textcolor{blue}{P}, \textcolor{orange}{D}, \textcolor{orange}{A}, \textcolor{blue}{H}, \textcolor{blue}{T}) = p(\textcolor{blue}{P})p(\textcolor{orange}{D})p(\textcolor{orange}{A} \mid \textcolor{blue}{P}, \textcolor{orange}{D})p(\textcolor{blue}{H} \mid \textcolor{blue}{P})p(\textcolor{blue}{T} \mid \textcolor{orange}{A}).$$

$$\begin{aligned} & p(P = 1, H = 1, T = 1) \\ &= \sum_a \sum_d p(P = 1) p(D = d) p(A = a \mid P = 1, D = d) && \text{(algorithm)} \\ &\quad \cdot p(H = 1 \mid P = 1) p(T = 1 \mid A = a) \\ &= p(H = 1 \mid P = 1) && \text{(constants out)} \\ &\quad \cdot p(P = 1) \sum_a p(T = 1 \mid A = a) \sum_d p(A = a \mid P = 1, D = d) p(D = d) \\ &= 0.9 \cdot 0.184564 && \text{((b), Step 2)} \\ &= 0.1661076. \end{aligned}$$

Step 3. Find the denominator: $G = \{H, T\}$, $G_{\text{ancestors}} = \{P, A, D\}$.



Joint distribution:

$$p(\textcolor{orange}{P}, \textcolor{orange}{D}, \textcolor{orange}{A}, \textcolor{blue}{H}, \textcolor{blue}{T}) = p(\textcolor{orange}{P})p(\textcolor{orange}{D})p(\textcolor{orange}{A} \mid \textcolor{orange}{P}, \textcolor{orange}{D})p(\textcolor{blue}{H} \mid \textcolor{orange}{P})p(\textcolor{blue}{T} \mid \textcolor{orange}{A}).$$

$$\begin{aligned} & p(H = 1, T = 1) \\ &= \sum_p \sum_a \sum_d p(P = p) p(D = d) p(A = a \mid P = p, D = d) && \text{(algorithm)} \\ &\quad \cdot p(H = 1 \mid P = p) p(T = 1 \mid A = a) \\ &= \sum_p p(H = 1 \mid P = p) && \text{(constants out)} \\ &\quad \cdot p(P = p) \sum_a p(T = 1 \mid A = a) \sum_d p(A = a \mid P = p, D = d) p(D = d) \\ &= 0.9 \cdot 0.184564 + 0.2 \cdot 0.53176 && \text{((b), Step 3)} \\ &= 0.1661076 + 0.106352 = 0.2724596. \end{aligned}$$

Step 4. Divide.

$$p(P = 1 \mid T = 1, H = 1) = \frac{0.1661076}{0.2724596} \approx 0.610.$$

Conclusion.

$$\underbrace{0.2}_{\text{prior}} \rightarrow \underbrace{0.529}_{H=1}, \underbrace{0.258}_{T=1}, \underbrace{0.610}_{H=1, T=1}.$$

An unhappy teacher is weak evidence: without the party, the teacher is also often unhappy (0.6647 in Step 3 of (b)). Two observations together give the strongest evidence.

3. Let $x_1 \sim \text{Poiss}(\lambda_1)$ and $x_2 \sim \text{Poiss}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$.

Idea. For $y = f(x_1, \dots, x_n)$ with discrete x_i , split the event $\{y = k\}$ into disjoint events and sum:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n) = \sum_{f(j_1, \dots, j_n) = k} \prod_i p(x_i = j_i) \quad (\text{if independent}).$$

Then simplify the sum and compare it with known distributions.

Solution. Goal: show that $p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^k / k!$ for all $k \geq 0$.

$$\begin{aligned} p(x_1 + x_2 = k) &= p\left(\bigcup_{j=0}^k \{x_1 = j, x_2 = k - j\}\right) && (x_1, x_2 \geq 0) \\ &= \sum_{j=0}^k p(x_1 = j, x_2 = k - j) && (\text{disjoint events}) \\ &= \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j) && (\text{independence}) \\ &= \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} \cdot e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!} && (\text{Poisson probabilities}) \\ &= e^{-\lambda_1} e^{-\lambda_2} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && (\text{factors without } j \text{ out of the sum}) \\ &= e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && (e^x e^y = e^{x+y}) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \frac{k!}{j! (k-j)!} \lambda_1^j \lambda_2^{k-j} && (\text{multiply and divide by } k!) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} && \left(\binom{k}{j} = \frac{k!}{j! (k-j)!}\right) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k && (\text{binomial theorem}). \end{aligned}$$

This is $\text{Poiss}(\lambda_1 + \lambda_2)$.

Conclusion. A sum of independent Poisson variables is a Poisson variable, and the parameters add. By induction, this is true for any finite number of variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. Find the maximum likelihood estimate of λ .

Idea.

$$L(\lambda) = \prod_i p(x_i | \lambda) \rightarrow \ell(\lambda) = \log L(\lambda) = \sum_i \log p(x_i | \lambda) \rightarrow \ell'(\lambda) = 0 \rightarrow \ell''(\lambda) < 0.$$

The logarithm is increasing, so L and ℓ have the same maximum point.

Solution. Let $S = \sum_{i=1}^N x_i$ and $\bar{x} = S/N$.

Step 1. Write the goal.

$$\begin{aligned} \hat{\lambda}_{\text{ML}} &= \arg \max_{\lambda} L(\lambda) \\ &= \arg \max_{\lambda} \ell(\lambda) \end{aligned} \quad (\log \text{ is increasing}).$$

Step 2. Find $\ell(\lambda)$ for Step 1.

$$\begin{aligned} \ell(\lambda) &= \log \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} && (\text{Poisson probabilities}) \\ &= \sum_{i=1}^N \log \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) && (\log \text{ of a product}) \\ &= \sum_{i=1}^N (-\lambda + x_i \log \lambda - \log x_i!) && (\log \text{ of each factor}) \\ &= -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i! && (\text{sum each term}). \end{aligned}$$

Step 3. Find the stationary point of ℓ .

$$\begin{aligned} \ell'(\lambda) &= -N + \frac{S}{\lambda}, \\ -N + \frac{S}{\lambda} &= 0 \implies \frac{S}{\lambda} = N \implies \lambda = \frac{S}{N} = \bar{x}. \end{aligned}$$

Step 4. Make sure that it is a maximum.

$$\ell''(\lambda) = -\frac{S}{\lambda^2} < 0 \quad \text{if } S > 0.$$

Thus ℓ is strictly concave, and $\bar{x}$ is the global maximum. If $S = 0$, then $\ell(\lambda) = -N\lambda$ decreases, and the maximum is at $\lambda = 0 = \bar{x}$.

$$\hat{\lambda}_{\text{ML}} = \bar{x}.$$

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda | a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda | x_1, \dots, x_N)$.

Idea. The numerator and the denominator of Bayes' theorem contain the same function:

$$p(\lambda | X) = \frac{p(X | \lambda) p(\lambda)}{\int p(X | \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

Do not calculate the integral directly:

$$\begin{aligned}
p(X | \lambda) p(\lambda) &= C \cdot g(\lambda) && (C \text{ does not depend on } \lambda) \\
g(\lambda) &= Z q(\lambda) && (q \text{ is a known density}) \\
\int CZ q(\lambda) d\lambda &= CZ && (\text{a density integrates to } 1) \\
p(\lambda | X) &= \frac{CZ q(\lambda)}{CZ} = q(\lambda).
\end{aligned}$$

The same method gives the posterior mean.

Solution. Let $S = \sum_{i=1}^N x_i$, $a' = a + S$, and $b' = b + N$.

Step 1. Write the goal.

$$\hat{\lambda}_B = \mathbb{E}[\lambda | X] = \int \lambda p(\lambda | X) d\lambda.$$

Step 2. Find $p(\lambda | X)$ for Step 1.

$$p(\lambda | X) = \frac{p(X | \lambda) p(\lambda)}{\int p(X | \lambda) p(\lambda) d\lambda}.$$

Step 2a. Separate the constant in the numerator of Step 2.

$$\begin{aligned}
p(X | \lambda) p(\lambda) &= \prod_{i=1}^N \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\text{likelihood and prior}) \\
&= \frac{\prod_i e^{-\lambda} \cdot \prod_i \lambda^{x_i}}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\text{split the product}) \\
&= \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\prod_i \lambda^{x_i} = \lambda^{\sum_i x_i}) \\
&= \frac{b^a}{\Gamma(a) \prod_i x_i!} \cdot \lambda^S \lambda^{a-1} \cdot e^{-N\lambda} e^{-b\lambda} && (\text{group the factors}) \\
&= \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a'-1} e^{-b'\lambda}}_{g(\lambda)} && (\text{add the exponents}).
\end{aligned}$$

Step 2b. Find the known density for g from Step 2a.

$$\mathcal{G}(\lambda | a', b') = \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} = \frac{b'^{a'}}{\Gamma(a')} g(\lambda) \implies g(\lambda) = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b').$$

Step 2c. Find the denominator of Step 2.

$$\begin{aligned}
\int CZ g(\lambda) d\lambda &= \int C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b') d\lambda && (\text{Step 2b}) \\
&= C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda | a', b') d\lambda && (\text{constants out of the integral}) \\
&= C \frac{\Gamma(a')}{b'^{a'}} && (\text{a density integrates to } 1).
\end{aligned}$$

Step 2d. Substitute Steps 2a–2c into Step 2.

$$\begin{aligned}
p(\lambda | X) &= \frac{Cg(\lambda)}{C \frac{\Gamma(a')}{b'^{a'}}} && (\text{Steps 2a, 2c}) \\
&= \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b')}{C \frac{\Gamma(a')}{b'^{a'}}} && (\text{Step 2b}) \\
&= \mathcal{G}(\lambda | a', b') && (\text{the constants cancel}) \\
&= \mathcal{G}(\lambda | a + S, b + N).
\end{aligned}$$

Step 3. Substitute Step 2d into Step 1.

$$\begin{aligned}
\mathbb{E}[\lambda \mid X] &= \int \lambda \mathcal{G}(\lambda \mid a', b') d\lambda && \text{(Step 2d)} \\
&= \int \lambda \cdot \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} d\lambda && \text{(gamma density)} \\
&= \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda && \text{(constants out of the integral).}
\end{aligned}$$

Step 3a. Find the integral for Step 3 with the same method. Use Step 2b with $a' + 1$ instead of a' :

$$\begin{aligned}
\int \lambda^{a'} e^{-b'\lambda} d\lambda &= \int \lambda^{(a'+1)-1} e^{-b'\lambda} d\lambda && (a' = (a' + 1) - 1) \\
&= \int \frac{\Gamma(a' + 1)}{b'^{a'+1}} \mathcal{G}(\lambda \mid a' + 1, b') d\lambda && \text{(Step 2b)} \\
&= \frac{\Gamma(a' + 1)}{b'^{a'+1}} \int \mathcal{G}(\lambda \mid a' + 1, b') d\lambda && \text{(constants out of the integral)} \\
&= \frac{\Gamma(a' + 1)}{b'^{a'+1}} && \text{(a density integrates to 1).}
\end{aligned}$$

Step 3b. Substitute Step 3a into Step 3.

$$\begin{aligned}
\mathbb{E}[\lambda \mid X] &= \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} && \text{(Step 3a)} \\
&= \frac{\Gamma(a' + 1)}{\Gamma(a') b'} && (b'^{a'}/b'^{a'+1} = 1/b') \\
&= \frac{a' \Gamma(a')}{\Gamma(a') b'} && (\Gamma(a' + 1) = a' \Gamma(a')) \\
&= \frac{a'}{b'} = \frac{a + S}{b + N}.
\end{aligned}$$

$$\hat{\lambda}_B = \frac{a + S}{b + N}.$$

Conclusion.

- *Conjugate prior:* gamma prior $\times$ Poisson likelihood $\rightarrow$ gamma posterior. The update only changes the parameters: $a \rightarrow a + S$ (events), $b \rightarrow b + N$ (observations).
- The estimate is a weighted mean of the prior mean and the estimate $\bar{x}$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{a}{b + N} + \frac{S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \bar{x} \xrightarrow{N \rightarrow \infty} \bar{x}.$$

- If $a, b \rightarrow 0$, then $p(\lambda) \propto \lambda^{a-1} e^{-b\lambda} \rightarrow 1/\lambda$, the non-informative prior from the notes below, and $\hat{\lambda}_B \rightarrow S/N = \bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).